

Project Report

CampusConnect Market

Abstract

CampusConnect Market is a web-based marketplace platform developed specifically for university campuses. The system enables students to buy and sell products within their campus community in a secure and convenient manner. It supports product listing, searching, cart management, wishlists, order placement, and role-based access control for customers, sellers, and administrators. The project demonstrates the practical implementation of Java Spring Boot, MySQL, HTML, CSS, and JavaScript to create a modern full-stack e-commerce platform.

Introduction

Traditional buying and selling of second-hand items among students often occurs through informal social media groups or word-of-mouth communication, making it difficult to verify sellers, manage transactions, and discover products efficiently.

CampusConnect Market addresses these challenges by providing a dedicated marketplace platform where students can securely trade products such as books, electronics, fashion items, and study materials. The platform allows users to browse products, communicate through a structured system, and manage purchases within a trusted campus environment.

Objectives

The primary objectives of CampusConnect Market are:

1. To provide a dedicated marketplace for campus communities.
 2. To enable students to buy and sell products easily.
 3. To support secure user authentication and authorization.
 4. To provide product search and category-based filtering.
 5. To facilitate cart and wishlist management.
 6. To enable efficient order placement and tracking.
 7. To provide seller dashboards for inventory management.
 8. To provide administrative tools for platform management.
 9. To encourage sustainable reuse of products among students.
 10. To demonstrate full-stack web application development.
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Technologies Used

Backend Technologies

- Java 17
- Spring Boot
- Spring Security
- Spring Data JPA
- JWT Authentication
- MySQL Database

Frontend Technologies

- HTML5
- CSS3
- JavaScript (ES6+)
- Fetch API
- Google Fonts

Development Tools

- Maven
- IntelliJ IDEA / Eclipse
- MySQL Workbench
- Git & GitHub

Methodology

The application follows a structured workflow:

1. Users register and log in securely using JWT authentication.
 2. Customers browse products available in the marketplace.
 3. Products can be searched and filtered by category and condition.
 4. Customers add desired products to their cart or wishlist.
 5. Orders are placed through a simple checkout process.
 6. Sellers manage product listings and inventory.
 7. Administrators monitor platform activities and users.
 8. All data is securely stored in a MySQL database.
 9. Dashboards provide insights into platform operations.
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Implementation Modules

1. User Authentication Module

- User Registration
- User Login
- JWT Authentication
- Role-Based Access Control

2. Product Management Module

- Product Listing
- Product Editing
- Product Deletion
- Category Management
- Product Condition Tracking

3. Search and Discovery Module

- Product Search
- Category Filtering
- Dynamic Product Catalog

4. Shopping Cart Module

- Add Products
- Remove Products
- Update Quantity
- Order Preparation

5. Wishlist Module

- Save Favorite Products
- Manage Wishlist Items

6. Order Management Module

- Checkout Process
- Order Tracking
- Purchase History

7. Seller Dashboard Module

- Product Management
- Sales Statistics
- Inventory Monitoring

8. Admin Dashboard Module

- User Management
- Product Moderation
- Category Administration
- Platform Monitoring

9. Customer Profile Module

- Personal Information Management
- Order History
- Account Settings

Results

Features Successfully Achieved

- Secure JWT-based authentication.
- Role-based user access control.
- Dynamic product catalog management.
- Product search and filtering.
- Shopping cart functionality.
- Wishlist management.
- Order processing system.
- Seller dashboard with sales insights.
- Admin dashboard for platform management.
- Responsive and modern user interface.

Sample User Flow

1. A student registers and logs into the platform.
2. The student browses available products.
3. Desired products are added to the cart.
4. The order is placed successfully.
5. The seller receives order details.
6. The administrator can monitor platform activity.

The system successfully achieves all project objectives and provides a complete campus marketplace solution.

Conclusion

CampusConnect Market provides a secure, efficient, and user-friendly marketplace solution for university campuses. The platform simplifies the buying and selling process while promoting sustainable product reuse among students. By integrating Spring Boot, MySQL, JWT authentication, and modern frontend

technologies, the project demonstrates the development of a scalable full-stack e-commerce application capable of serving campus communities effectively.

Future Scope

The following enhancements can be implemented in future versions:

1. Online payment gateway integration.
2. Real-time chat between buyers and sellers.
3. Product recommendation system using AI.
4. Mobile application development.
5. Push notifications and email alerts.
6. Product rating and review analytics.
7. Campus verification using student IDs.
8. Location-based product discovery.
9. Multi-campus marketplace support.
10. Advanced sales and business analytics.
11. AI-powered fraud detection.
12. Cloud deployment and scalability improvements.